Dingli Ma

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Education

University of Washington — Seattle, USA
Ph.D. Candidate in Operations Management — Sep 2023 – Present
Advisors: Prof. Michael R. Wagner

Columbia University — New York, USA
M.S. in Operations Research — Sep 2021 – Feb 2023

Fudan University — Shanghai, China
B.S. in Mathematics and Applied Mathematics — Sep 2017 – Jun 2021

Publications

Journal

1. Liao, L., Fu, Z., Yang, Z., Wang, Y., **Ma, D.**, Kolar, M., Wang, Z. (2024). *Instrumental Variable Value Iteration for Causal Offline Reinforcement Learning.* **Journal of Machine Learning Research**, 25(303), 1-56.
   - Identified Markov transition dynamics from observational data via instrumental variables; proposed Instrumental Variable Value Iteration (IVVI) algorithm.
   - Implemented IVVI algorithm in linear, nonlinear settings and newly built movie recommender system; provided consistency and empirical validation.

Research In Progress

1. **Ma, D.**, Zhang, M., Chen, S., Luo, H., Wang, Y. *Contextual Non-Perishable Inventory Control with Censored Demand and General Function Approximation.* Work in progress.
   - Presented at INFORMS Workshop on Data Science 2025.
   - Proposed a lazy-update inventory control policy (Binary Search with an Offline Oracle) that integrates offline regression oracle and online learning to address censored demand.
   - Established optimal regret under realizability assumption; controlled implementability via infrequent policy updates.
   - Implemented simulations and real-data experiments demonstrating superior cost and stability across diverse contextual demand scenarios.
2. **Ma, D.**, Wagner, M. *Delivering on the Two-Sided Promise Window via Crowd-Courier Logistics.* Work in progress.
   - Formulated a robust staffing/routing with *two-sided* delivery windows under uncertainty in arrivals, preparation, travel, and on-site service.
   - Derived tight worst-/best-case service-time bounds and a tractable deterministic counterpart for wage, staffing, and window co-design.

3. **Ma, D.**, Wang, J., et al. *Scientific Misinformation Detection: Fact-Checking in Complex Scenarios.* Work in progress.
   - Constructed a scientific misinformation dataset by collecting and structuring claim-level biomedical examples and associated entity-centric knowledge graphs.
   - Benchmarked LLMs on the dataset and explored reinforcement learning methods to improve open-source LLM agents for evidence retrieval and scientific misinformation classification.

Professional Experience

Intern, AGV Department — Eoslift Corporation, Shanghai, China　　Mar 2021–Jun 2021

- Collaborated with AGV team on improving optimization-based collision avoidance algorithm by creating a new smoothening planned path method, increasing algorithm efficiency by 10%; worked with a group to build a simple simulation by using MATLAB.
- Examined path planning and collision avoidance algorithm with people from Engineering Department in a real factory; assisted in installation of communication module on 4 Automated Guided Vehicles (AGVs).

Data Analysis Intern, Big Data Team — Nielsen, Remote　　May 2020–Jul 2020

- Updated and analyzed weekly and monthly data report of Volvo Cars in China and provided data analysis solutions; communicated and worked with colleagues about how to present data from social media.
- Drafted a 30-page industrial survey report under guidance of supervisor, including client demand analysis and market analysis.

Teaching Experience

University of Washington (Teaching Assistant)
QMETH 500: Statistical Data Analysis For Management (HMBA), Winter 2026;
QMETH 505: Decision Modeling (HMBA), Spring 2026;
OPMGT 565: Business Analytics—Tools for Big Data (MBA), Spring 2025; Spring 2026
SCM 512: Spreadsheet Modeling for Business Enterprise (MSCM), Summer 2024
QMETH 201: Introduction to Statistical Methods (Undergraduate), Winter 2024; Spring 2024

Columbia University (Course Assistant)
IEOR 4004: Optimization Models and Methods, Fall 2022

Skills

Programming Languages: Python, MATLAB, R
Methodologies: Reinforcement Learning, Multi-armed Bandit, Online Learning, Robust Optimization, Stochastic Optimization
Tools: GitHub, LaTeX